COMPOUND PP

Glass Fiber Modified PP Compound Resins

- ►GH23S
- ►GH41
- ►GH42
- ►GH42M
- ►GH43

Description

G/F modified Compound PP is modified by glass fiber filling and to improve mechanical rigidity and heat-resistance. It is produced made by modifying . A variety of base PPs, including HIPP (High Isotactic Polypropylene), with Hanwha Total's special processing technology.
Its excellent quality makes it ideal for use in air conditioner fans and automobile cooling fans; and like products requiring high rigidity and high impact-resistance.

Characteristics

There is a wide selection of Hanwha G/F PP products available for uses in products that require high- and ultra-high rigidity and high flowability. Adjustments depending upon molding conditions present no difficulty. Several Hanwha Total G/F PPs have improved dimensional stability thanks to blends with mica, and fire-resistance ratings of UL94 V-0.

Applications

G/F PP helps improve rigidity and impact-resistance simultaneously dimensional stability, while providing heightened heat-resistance.
These grades are high-temperature strengths with added .
The most common uses are in:

- Automobile parts, including brackets, switch boxes, cooling fans and oil caps;
- Electric and electronic parts, such as air conditioner fans, washing machine drums, pulley/legs for washing machines, and more;
- Household/industrial appliances, parts for electric tools, showers, bidets , etc.

Major Property Requirements

- Excellent mechanical strength and heat-resistance;
- Low warpage
- Processability

Main Property of Mica Modified PP

Base Polymer Homo-PP

- Mechanical strength,

G/F C ompound PP has 2~3-times higher tensile strength, and 4-times better flexural strength, than other general PP.
Figure 1 shows the change in the physical reinforcing characteristics of G/F Compound PP when glass fiber is filled.

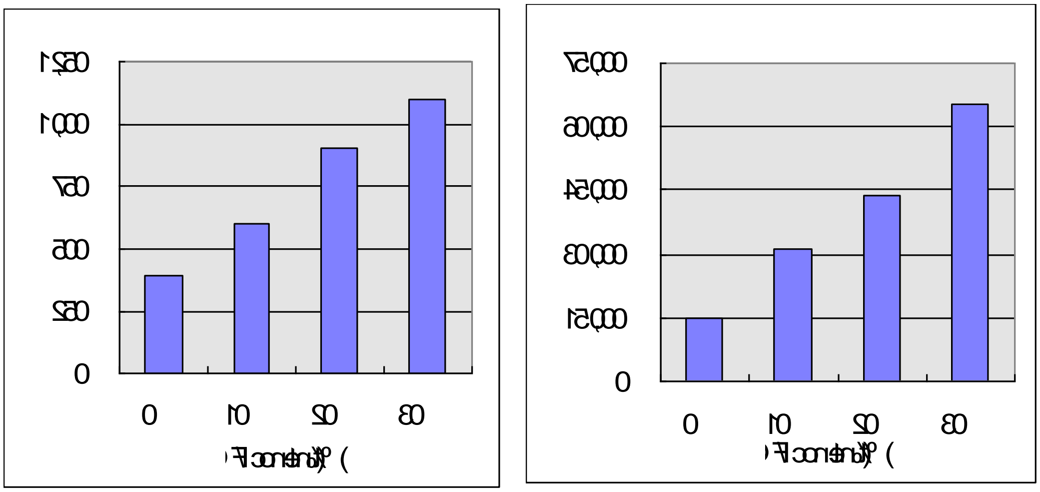

Figure 1. Comparisons of reinforced physical properties among mica-modified PP; mica and G/F-modified PP; and talc-modified PP.

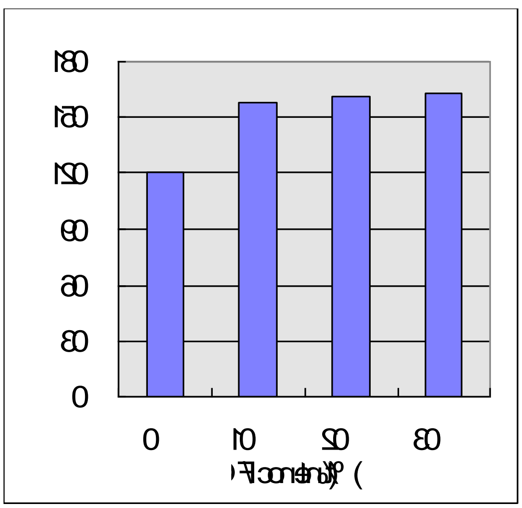

Figure 2. Heat-resistance with Glass Fiber Filling

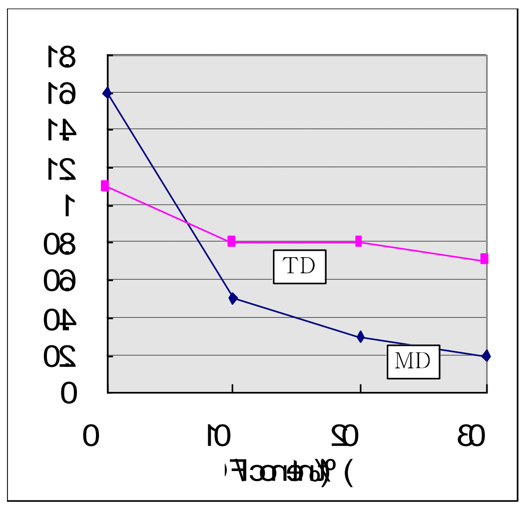

Figure 3. Shrinkage Rate with Glass Fiber Filling

▶ High Heat-Resistance,Hanwha G/F Compound PP exhibits high temperatures thermal deformation as the glass fiber filler protects the base PP from deformation. Thermal deformation temperatures are about 40~50℃ higher than those of non- modified common PP. (See Figure 2.)

▶ Fire-Resistance: General G/F PP have UL94 ratings, while flame-retardant G/F PP is rated at UL94 V-0

▶ Excellent Molding Characteristics: Hanwha G/F Compound PP has a wider molding range and lower molding temperature (20~30℃), than other G/F modified resins.

▶ Low Shrinkage Rate: Hanwha G/F Compound PP has a significantly lower shrinkage rate than common PPs. However, shrinkage-rate anisotropy, due to G/F direction, can occur during molding. This must be carefully considered at the design stage. As seen in Figure 3, the Machine Direction, MD, shrinkage rate sharply drops as G/F content increases; while the Transverse Direction, TD, decreases less sharply.

● General Processing Guide

G/F Compound PP injection conditions are similar to the general conditions of other PP with excellent processability. Typical processing conditions for injection are as follows:

Conditions	GH23S,GH42M	GH41, GH42, GH43
Cylinder Temp.(℃)	200 ~ 240	180 ~ 220
Mold Temp.(℃)	50 ~ 80	50 ~ 80
Screw RPM	30 ~ 80	30 ~ 80
Injection Pressure(kgf/cm^2)	600 ~ 1,100	500 ~ 900
Injection Speed	max	max
Back Pressure(kgf/cm^2)	5 ~ 10	5 ~ 10

* In general, pre-drying is unnecessary, and the injection pressure is slightly increased over those in common PPs based on G/F content.
The use of scrap to reduce costs must be avoided because it can cut G/F.

Physical Properties

Resin Properties

Properties	Test Method	Unit	GH41	GH42	GH43
Physical Property					
Melt Index	ASTM D1238 230℃	g/10min	12	10	7.0
Specific Gravity	ASTM D792	g/cm^3	0.97	1.04	1.13
Mechanical & Thermal Property					
Tensile Strength	ASTM D638 50mm/min	kg/cm^2	600	900	1,100
Elongation at Break		%	5	4	3
Flexural Stiffness	ASTM D790 5mm/min	kg/cm^2	850	1,100	1,500
Flexural Modulus			28,000	42,000	60,000
Izod Impact Strength	ASTM D256 23℃	kg.cm/cm	8.0	10.0	13.0
Heat Deflection Temp.	ASTM D648 4.6kg/㎠	℃	160	162	163
Rockwell Hardness	ASTM D785	R-Scale	113	114	115
Processing					
Shrinkage Ratio	HANWHA TOTAL	%	0.4~1.1	0.3~1.0	0.2~0.9
Spiral Flow	HTC				
Certifications					
UL94	-		HB	HB	HB

* Data shown above are representative values for referance purposes only, and not to be construed as specifications.

▶ **Long-term heat resistance Grades**

Properties	Test Method	Unit	GH23S	GH42M
Physical Property				
Melt Index	ASTM D1238 230℃	g/10min	3.0	5.0
Specific Gravity	ASTM D792	g/cm^3	1.12	1.04
Mechanical & Thermal Property				
Tensile Strength	ASTM D638 50mm/min	kg/cm^2	1,000	900
Elongation at Break		%	10	3
Flexural Strength	ASTM D747 5mm/min	kg/cm^2	1,350	1,100
Flexural Modulus	ASTM D790 5mm/min		60,000	45,000
Izod Impact Strength	ASTM D256 23℃	kg.cm/cm	12.0	8.0
Heat Deflection Temp.	ASTM D648 4.6kg/㎠	℃	162	155
Rockwell Hardness	ASTM D785	R-Scale	110	110
Processing				
Shrinkage Ratio	HANWHA TOTAL	%	0.2~0.9	0.3~1.0
Spiral Flow	HANWHA TOTAL			
Certifications				
UL94	-		HB	HB

● Others

▶ Package units: Woven bags of 25 and 500kg.

Food Contact Application

- There may be some limitation to apply Hanwha Total GH23S, GH41, GH42, GH42M, GH43 to the food packaging
- Thus, the verification on the suitability is necessary. In case you might need additional information, please contact Hanwha Total Composite Development Team.

Other Information

The information in this document can be used for reference only, not to be construed as specification. Customers are responsible for determine whether our product and information is suitable for their particular purpose and for the compliance with related law.
HANWHA TOTAL assumes no obligation or liability for the information in this document.